01. Multimodal Data Ingestion & Storage

Stream 01: Static EHR
- 60 Clinical Baseline Features
- Spirometry: FEV1%, FEV1/FVC, PEF
- Prescribed Controller Regimen & ICU History
- Location: aasthma/dataset/data/static/

Stream 02: Wearable Telemetry
- 20 Continuous Channels (12h Lookback Window)
- SpO2, Heart Rate, HRV (rMSSD), Respiratory Rate
- Nocturnal Awakenings & SABA Inhaler Puffs
- Cough, Wheeze, Chest Tightness, Dyspnea Scores
- Location: aasthma/dataset/data/dynamic/

Persists

Persists

PostgreSQL 16 Relational Storage
- patients & patient_history tables
- timeseries_telemetry (indexed by hour)
- clinical_events & severity labels
- patient_daily_logs (PRO check-ins)
- user_accounts (Doctor & Patient RBAC)
- Location: aasthma/database/models.py

Persists

Stream 03: Environment (AQI)
- Ambient PM2.5, PM10, AQI Index
- 70+ Indian CPCB Monitoring Stations & Open-Meteo
- Simulation Replay Clock (t = 12h to 239h)
- Location: aasthma/simulation/clock.py

Static EHR (60D)

Telemetry (20ch)

Wearable Vitals

Ambient PM2.5 Dose

02. Hybrid Digital Twin Engine

Stage 1: Base Predictors

Stage 1A: Dilated TCN Fusion Model
- StaticMLPEncoder (60D -> 64D Embedding)
- Conv1DDynamicEncoder (20 channels x 12h)
- 3 Dilated Conv1D Blocks (d = 1, 2, 4)
- Multi-Head: P(24h), P(48h), Severity (Grade 0 to 3)
- Artifact: aasthma/training/model1/digital_twin_model.pt

Stage 1B: XGBoost Multi-Target Suite
- Tabular Gradient-Boosted Decision Trees
- Rolling Window Statistics (mean, std, min, max, delta)
- Temporal Lag Shifts (t-2, t-1, t, t+1, t+2)
- Multi-Head: P(24h), P(48h), Severity (Grade 0 to 3)
- Artifact: aasthma/training/model2/gb_digital_twin_models.pkl

Mechanistic Airway Simulator (Latent ODE)
- dI/dt = -k*I(t) + a*Dose_PM(t) + b*Trigger(t) + d_sys - c*Controller(t) + sigma*xi(t)
- Euler-Maruyama Numerical ODE Integration (dt = 1.0h)
- Continuous Airway Mucosal Inflammation State: I(t) in [0.05, 1.0]
- Counterfactual What-If Sandbox: Stay Indoors, N95 Mask, HEPA, ICS Step-Up
- Re-scores Counterfactual Trajectories with Latency < 65 ms
- Location: aasthma/simulation/mechanistic_simulator.py

6D TCN Probs

6D XGB Probs

Counterfactual Rescoring

ODE Modifiers

Stage 2: Stacking MLP Meta-Learner
- RespiTwinMLPMetaLearner (12D -> 32D -> 16D -> 3 Output Heads)
- Calibrates 12D Probabilities: [tcn_24h, xgb_24h, tcn_48h, xgb_48h, tcn_sev_0..3, xgb_sev_0..3]
- 90% Conformal Risk Prediction Intervals & SHAP Feature Attributions
- **Held-Out Test Results: 24h ROC-AUC = 0.940 | 48h ROC-AUC = 0.943 | Severity F1 = 0.844**
- Location: aasthma/training/ensemble.py (meta_ensemble_model.pt)

EWC Retraining

Stage 3: LLM Clinical Extractor & Retraining Pipeline
- Converts free-text clinical notes (Typhoid, Dengue, Dehydration) into ODE & vital deltas
- Doctor-verified clinical outliers saved into PostgreSQL Experience Replay Buffer
- Continual Retraining: Elastic Weight Consolidation (EWC) updates Meta-Learner without forgetting
- Generates Pydantic-validated Copilot Reports (precautions, triggers, breathing exercises)
- Location: aasthma/llm/clinical_extractor.py & aasthma/llm/llm.py

Inflammation State I(t)

Calibrated Forecast

Copilot Reports

03. Service Gateway & User Interfaces

FastAPI Service Layer (:8000)
- REST Endpoints: /api/patients, /api/what-if, /api/clock
- Note Adaptation & Continual Retrain: /api/patients/{id}/clinical-note, /api/model/adapt-outliers
- Copilot Report Delivery: /api/patients/{id}/copilot-report
- PostgreSQL Dual-Role RBAC Authentication: /api/auth/login, signup, verify
- Location: aasthma/server.py

Clinician Data Stream

Patient Data Stream

Next.js 16 Web Applications (:3000)

Treating Doctor Portal
- /patients: Cohort Risk Triage Roster & 240h Simulation Replay Clock
- /patients/[id]: 3D GLB Anatomical Lung & GLSL Thermal Inflammation Viewer
- /patients/[id]/forecast: 24h & 48h Conformal Fan Chart & SHAP Drivers
- /patients/[id]/what-if: Interactive Counterfactual Intervention Sandbox
- /patients/[id]/adaptation: Free-Text Note Parser & EWC Model Retraining
- /benchmarks & /assumptions: Physiological Ablations & Auditable Citations

Asthma Patient Portal
- /patient-portal: Personal 24h & 48h Risk Guidance Card
- Real-Time Local CPCB Station AQI & Smog Inversion Advisories
- Daily Self-Reported Outcomes (PRO): SABA Puffs & Symptom Check-ins
- Clinician-Verified Copilot Report Access (Precautions & Breathing Plans)
- Role-Isolated Patient Authentication (PT-8821, PT-4409, PT-1923)